

Physiology of the digestive system



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Gastric motility and emptying



Roadmap

- Control of motility
- Emptying into the small intestines
- The process of vomiting



Mastication

- ✓ Physical breakdown

Swallowing

- ✓ Oral, pharyngeal and esophageal phases

Peristalsis

- ✓ Unidirectional wave-like smooth muscle contractions

Filling Phase

- ✓ Receptive relaxation: smooth muscle reduces tone in response to food
- ✓ Stomach Volume:
~0.5L when empty
~0.8L to 4.0L during receptive relaxation

Salivary glands:

Parotid

Sublingual

Submandibular

Tongue

Bolus

Pharynx

Epiglottis

Trachea

Relaxed esophageal sphincter

Gastroesophageal sphincter

Fundus

Pyloric sphincter

Duodenum

Body

Antrum



ANS Comparison

Sympathetic branch

- Uses Noradrenaline
- Decreases motility
- Decreases secretion
- Contracting sphincters

Parasympathetic branch

- Uses Ach
- Increases motility
- Increases secretion
- Relaxing sphincters



Motility of the stomach which leads to mixing of the food with gastric juice and the subsequent emptying of stomach contents into the small intestines, is maintained locally by intrinsic nervous collection under the muscle layer



- Distension of the stomach by food causes a peristaltic wave, that to travel from the greater curvature towards lesser curvature and pyloric antrum at a rate of 3 contractions per minute.
- The contractions increase with the hour



- At first, these waves are electrical and not yet contractions for the action potential threshold is not reached
- However, with acetylcholine released by Vagus nerve and/or Gastrin hormone the threshold is reached and an action potential ensues



The local control within the wall

2 plexuses are imbedded in the wall of tract:

- The submucosal plexus: Meissner's plexus

It is found in submucosal layer to control GI **secretion**

- The myenteric plexus: Auerbach's plexus

It is found between the 2 muscular layers to control the GI motility (**contraction**)



- Gastrin constricts lower esophageal sphincter but relaxes pyloric sphincter, so does not cause an increase in intragastric pressure.
- Acetylcholine increases the length of the muscle fiber i.e. increase size of stomach.
- Therefore, the Vagus nerve allows the stomach to accept food without **changing intragastric pressure** (increase in size) and also allows mixing waves of contraction, **(receptive relaxation)**



What determines gastric emptying?

- The rate at which the stomach empties its contents into the duodenum depends upon:

Factors related to the state of the muscles...

1. Contractions of the stomach
2. State of pyloric sphincter (closed or open)
3. Tone of the duodenal wall



Factors related to the food (amount and consistency)

- Liquids depart the stomach more rapidly
- Large volumes are released more rapidly than small volumes.
- Acidity of duodenal contents delay gastric emptying. If duodenal pH goes below 4 gastric emptying stops until the contents are neutralized by the alkaline pancreatic juice and bile.



Factors related to the food (amount and consistency)

- Osmolality of duodenal contents: emptying of gastric contents is fastest when the Osmolality of chyme is about 200 milliosmole per liter. At higher or lower osmolality emptying is delayed.
- Presence of fat and their digestive products in duodenum delay the rate of emptying.



Other factors

- Duodenal distension delays gastric emptying by stimulating sympathetic nervous system.
- Emotional state such as fear delays emptying probably thorough sympathetic nerves.
- Cold fluids delay gastric emptying compared to fluids at 37OC.



Gastric emptying

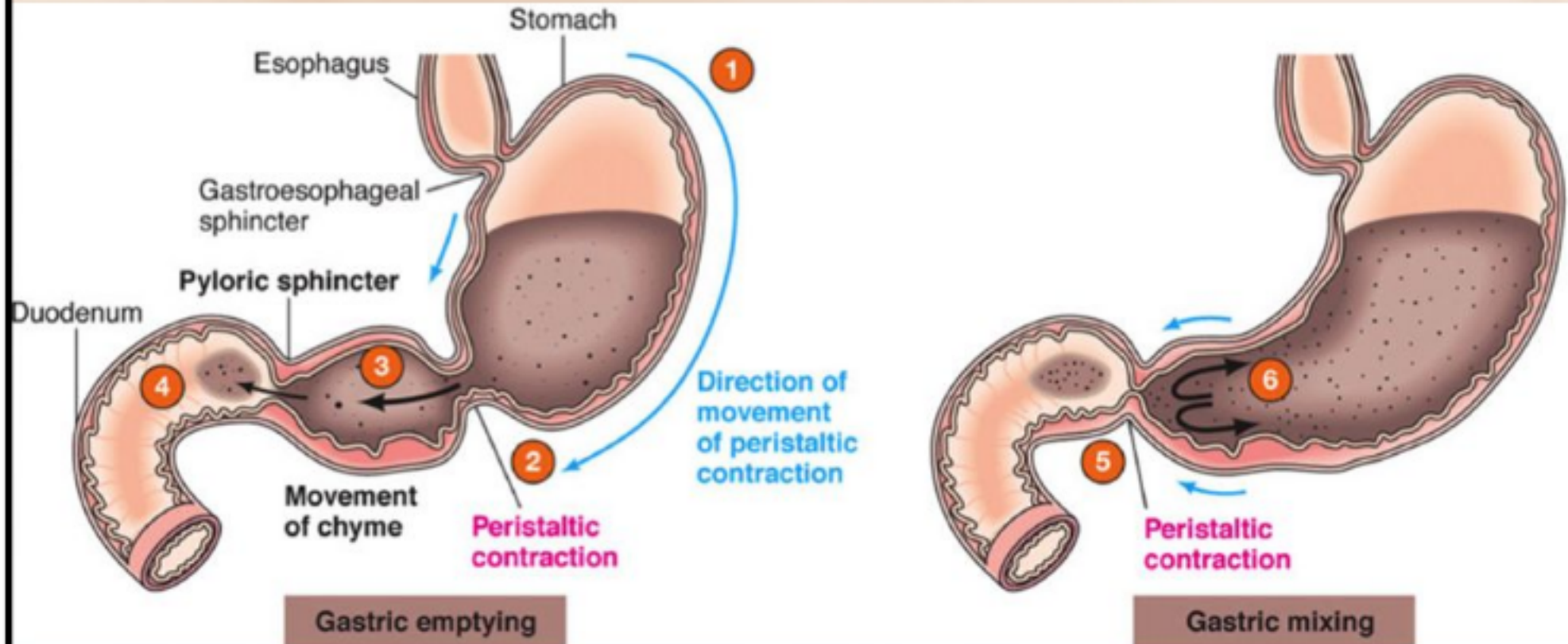
- The antrum, pylorus and upper duodenum function as a unit
- The pyloric sphincter:- is an atypical sphincter

When the peristalsis reaches the lower part of stomach the antrum, pylorus and proximal duodenum are relaxed

- Chyme in liquid form passes to duodenum being pushed along by tonic contraction of proximal stomach



Process of Gastric Emptying



- 1) Gastric peristalsis
- 3) Antral peristalsis

- 2) Vigorous antral contraction
- 4) Relaxation of pylorus



- The terminal antrum then contract aiding the propulsion of liquid chyme
- This followed by contraction of the pylorus, which closes off the stomach and arrests emptying to allow grinding of solids in chyme
- The proximal duodenum contracts to moving chyme to distal duodenum



Vomiting

- It is the forceful evacuation of gastric material through the mouth and/or nose
- a protective mechanism by which the stomach removes harmful materials
- Substances that induce vomiting are called emetics
- Irritation or over distension of the stomach or duodenum is the strongest stimulus



Causes of vomiting:

- In most cases nausea precedes vomiting .
1. Irritation of pharynx
 2. Irritation of stomach & intestines
 3. Disturbances of semicircular canals
 4. Early pregnancy
 5. Drugs



- vestibular system:- by motion sickness
- Frontal lobe: by psychic stimuli (shocking sights or smells)
- The chemoreceptor trigger zone : directly stimulate vomiting center (morphine, apomorphine, digitalis derivatives and tartar emetic)



Pathway

- Receptors: are situated in pharynx, stomach, duodenum
- Afferents: from the stomach & duodenum : vagus , sympathetic nerves
- Vomiting center: a collection of neurons present bilaterally in the medulla oblongata
- Efferents: cranial nerves (V,VII, IX, X,XII), spinal nerves to diaphragm and abdominal muscles
- Effectors:
Abdominal muscles, diaphragm, intestine, stomach, esophagus & mouth



The mechanism of vomiting

- Vomiting start with closure of the glottis & nasopharynx
- Violent contraction of abdominal muscles & diaphragm to squeeze the stomach and increase its pressure
- The wall of stomach relaxes, the LES opens and the UES suddenly open
- The high intra-gastric pressure allows gastric contents to expelled through esophagus to the outside



Till next lecture

